

Be the Recognizer

Today **you** are the voice assistant. Can you beat the noise?

Name: _____

Partner: _____

1 • Partner up

The Speaker holds the sentence cards. The Recognizer must not peek.

2 • Round 1: quiet

The Speaker reads cards 1–3 **once** each, in a normal voice.

3 • Round 2: noisy

Music plays. The Speaker **whispers** cards 4–6, once each.

4 • Score it

Compare each card with what you wrote. Count your errors below.

Write exactly what you hear, even if it makes no sense

#	WHAT I HEARD	WORDS SAID (N)	SWAPPED (S)	MISSING (D)	EXTRA (I)	ERRORS S+D+I
ROUND 1 • QUIET ROOM, NORMAL VOICE						
1						
2						
3						
Round 1 totals		21				
ROUND 2 • MUSIC ON, WHISPERED						
4						
5						
6						
Round 2 totals		21				
BONUS • SWAP ROLES IF THERE IS TIME						
7						
8						

Word Error Rate = (S + D + I) ÷ N

S = swapped: a wrong word in place of a right one

I = extra: a word written that was never said

Round 1 WER: _____ %

D = missing: a word that was said but not written

N = the number of words the Speaker said

Round 2 WER: _____ %

Worked example

Said: call mom at four (N = 4)

Heard: call tom at for the mom→tom and four→for are 2 swaps (S = 2). "the" is extra (I = 1). Nothing is missing (D = 0).

WER = (2 + 0 + 1) ÷ 4 = 75%

Scoring: ignore capitals and punctuation. "4" and "four" match; "for" and "four" do not.

Think about it

1. Which round had the higher WER? What did the noise and whispering take away?

2. When did the other words in the sentence help you guess right (or trick you)?

Speaker Cards

Cut apart (or fold) and keep them hidden from the Recognizer.

Speaker rules: read each card **one time**. No repeats, no spelling, no hints. A real voice assistant gets one try, too. After each round, show your cards so you can score together and tell the Recognizer how many words each card has (N).

ROUND 1 · CARD 1

Quiet room. Say it once, in a normal voice.

“Please call Mom at four.”

N = 5 words

ROUND 1 · CARD 3

Quiet room. Say it once, in a normal voice.

“Set a timer for fifteen minutes.”

N = 6 words

ROUND 2 · CARD 5

Music on. Whisper it once.

“Add eight pears to the shopping list.”

N = 7 words

BONUS · CARD 7

Swap roles. Pick quiet or noisy.

“The stuffy nose can lead to problems.”

N = 7 words

ROUND 1 · CARD 2

Quiet room. Say it once, in a normal voice.

“I scream, you scream, we all scream for ice cream.”

N = 10 words

ROUND 2 · CARD 4

Music on. Whisper it once.

“It is hard to recognize speech.”

N = 6 words

ROUND 2 · CARD 6

Music on. Whisper it once.

“Text Siobhan that the game starts at two.”

N = 8 words

Say it: **shi-VAWN**

BONUS · CARD 8

Swap roles. Pick quiet or noisy.

“Meet me at the park on the thirteenth.”

N = 8 words

✂ cut along the dashed lines

Be the Recognizer

Students experience the core problem of speech recognition, then measure it the way engineers do.

TIME

15 minutes

GRADES

4–12 (adaptations below)

GROUPS

Pairs, or trios with a scorekeeper

MATERIALS

Pages 1–2, pencils, a phone or speaker for music

Run of show

- 0:00 Hook (2 min).** “Has a voice assistant ever misheard you?” Today, students are the voice assistant.
- 0:02 Round 1 (3 min).** Quiet room. Speakers read cards 1–3 once, in a normal voice. Recognizers write exactly what they hear.
- 0:05 Round 2 (3 min).** Play music at a moderate volume. Speakers whisper cards 4–6 once.
- 0:08 Score (4 min).** Pairs compare each card with what was written and count S, D and I. Collect each pair’s round totals on the board. Class WER = all errors ÷ all words, for each round.
- 0:12 Discuss (3 min).** Use the prompts below. Bonus cards are for early finishers.

Scoring notes

- Ignore capital letters and punctuation.
- “4” and “four” count as the same word. Engineers call this *text normalization* and apply it before computing WER.
- Sound-alikes with a different spelling (for/four, pears/pairs, two/to) are errors: a machine that writes the wrong word may do the wrong thing.
- WER can exceed 100% when many extra words are written.

Adaptations and extensions

- Younger students:** skip the formula and count “oops words” per round. **Grades 9–12:** compute WER as a percentage, explain why it can exceed 100%, and graph class results by round.
- Human vs. machine:** read the same cards into the voice typing on a phone or Chromebook, score its WER, and compare with the class. Many online speech tools send audio to the cloud, so follow your school’s technology and privacy policy.
- AI4K12 connection:** Big Idea 1 (Perception) and Big Idea 4 (Natural Interaction) from the Five Big Ideas in AI (ai4k12.org).

What each card tests

1 Sound-alikes: Mom / Tom, four / for

2 Word boundaries: “I scream” / “ice cream”

3 Numbers that sound alike: fifteen / fifty

4 Word boundaries: “recognize speech” / “wreck a nice beach”

5 Homophones: eight / ate, pears / pairs

6 A rare name (Siobhan) plus two / to / too

7 Word boundaries: “stuffy nose” / “stuff he knows”

8 Numbers that sound alike: thirteenth / thirtieth

Discussion prompts

- Which round had the higher class WER? What did the noise and whispering take away?
- “I scream” or “ice cream”? The sounds are nearly the same. How did you decide?
- Why was “Siobhan” so hard? You have probably heard that name far less often than “Sarah”.
- Who might voice assistants mishear most often? How could engineers fix that?

The science connection

- Speech recognizers are graded with this exact metric: word error rate, $WER = (S + D + I) \div N$.
- A recognizer combines an **acoustic model** (which sounds did I hear?) with a **language model** (which words usually go together?). Guessing “ice cream” from context is the language model at work.
- Noise and whispering hide acoustic detail. Engineers train models on audio mixed with noise so they learn to cope.
- Rare words and names remain hard for people and machines alike, and improving them is an active research area.